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GitHub Link	https://github.com/saimounika1315/MLA0403---Deep-Learning

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Design and Develop Model

COURSE NAME: Deep Learning
SLOT : A

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10%

Assessment Tool Weightage: 30%

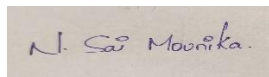
Duration: 90 minutes

Mark : 25

Code of Conduct:

I **Narra Sai Mounika / 192425367** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Narra Sai Mounika
Name of the student:

Criteria	Article Selection & Problem Understanding (2)	Deep Learning Model & Methodology Analysis (2.5)	Results & Performance Evaluation (2.5)	Critical Analysis & Technical Evaluation (2)	Future Scope, Presentation & Documentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						
Total						

QUESTIONS: 01

Total Marks:25

Select and review a recent research article related to the below **topic**, such as **Transfer Learning, DenseNet, PixelNet, RNN, Bidirectional RNN, Encoder–Decoder/Seq2Seq, BPTT, or LSTM**. Summarize the **problem addressed, dataset used, proposed deep learning model, methodology, performance metrics, major results, and key findings**, and critically comment on the **advantages, limitations, and possible future improvements** of the approach presented in the article.

Prepare a structured Article Review addressing the following sections:

1. Article Details

Provide the following information:

Title of the research article

Names of the authors

Journal/Conference name

Year of publication

DOI or publication link

Topic/Deep Learning technique addressed

Answer:

Title: Deep Residual Learning for Image Recognition

Authors: Kaiming He, Xiangyu Zhang, Shaoqing Ren, Jian Sun

Conference: IEEE Conference on Computer Vision and Pattern Recognition (CVPR)

Year of Publication: 2016

Publication Link:

[Research Paper \(CVPR 2016\)](#)

Topic / Deep Learning Technique Addressed:

Transfer Learning using ResNet (Residual Neural Network)

2. Research Problem

What problem does the article attempt to solve?

Why is the problem important?

What are the limitations of existing approaches identified by the authors?

Answer:

Problem Addressed

The research focuses on improving image classification accuracy in very deep neural networks.

Importance of the Problem

- Deep networks generally provide better feature extraction.
- Increasing network depth often leads to degradation problems.
- Training becomes difficult due to vanishing gradients.

Limitations of Existing Methods

- Traditional CNNs showed reduced accuracy when depth increased.
- Difficulty in training very deep architectures.
- Poor gradient propagation.

3. Objectives of the Study

What are the main objectives of the research?

What specific goals does the proposed model attempt to achieve?

Answer:

Main Objectives

1. Develop extremely deep neural networks.
2. Overcome degradation and vanishing gradient problems.
3. Improve image classification performance.

Specific Goals

- Introduce residual learning.
- Achieve higher accuracy than existing CNN models.
- Enable efficient transfer learning.

4. Dataset Description

What dataset(s) are used?

What is the size of the dataset?

What type of data is used—images, text, time-series, video, etc.?

How is the dataset divided into training, validation, and testing sets?

Answer:

Dataset Used

ImageNet Large Scale Visual Recognition Challenge (ILSVRC 2015)

Dataset Size

- More than 1.2 million training images
- 50,000 validation images
- 1000 object categories

Type of Data

- Image Dataset

Data Split

Dataset Images

Training 1.2 Million

Validation 50,000

Testing Official ImageNet Test Set

5. Proposed Methodology

Explain the overall methodology adopted in the article.

Identify the deep learning architecture used.

Explain the important layers/components of the model.

Describe the data preprocessing techniques used.

Explain the training procedure and important parameters, wherever available.

Answer:

Overall Methodology

Input Image



Preprocessing



Residual Blocks



Feature Extraction



Global Average Pooling



Fully Connected Layer



Softmax Output

Deep Learning Architecture

- ResNet-34
- ResNet-50
- ResNet-101
- ResNet-152

Important Components

Residual Block

Input (x)



Convolution



Convolution



Add Skip Connection



Output $F(x)+x$

Data Preprocessing

- Image resizing
- Normalization
- Data augmentation
- Random cropping

Training Parameters

- SGD Optimizer
- Batch Normalization
- ReLU Activation
- Cross Entropy Loss

6. Deep Learning Technique Used

Explain how the selected technique is applied in the research.

How is Transfer Learning used?

How does DenseNet perform feature reuse?

How is RNN/LSTM used to process sequential data?

How does a Bidirectional RNN utilize information from both directions?

How does an Encoder–Decoder/Seq2Seq architecture process input and output sequences?

How is BPTT used for training recurrent networks?

Answer:

Transfer Learning

Transfer learning utilizes knowledge learned from ImageNet and applies it to smaller datasets.

Process

Pre-trained ResNet



Freeze Early Layers



Replace Final Layer



Fine Tune Last Layers



Target Task

Benefits

- Reduced training time
- Better accuracy
- Less data requirement

7. Model Architecture

Draw or reproduce a simplified architecture diagram of the proposed model and explain:

Input → Preprocessing → Feature Extraction/Encoder → Hidden Layers →

Classification/Decoder → Output

Identify the role of the major components in the architecture.

Answer:

Simplified Architecture

Input Image



Image Preprocessing



Conv Layer



Residual Block 1



Residual Block 2



Residual Block 3



Residual Block 4



Global Average Pooling



Fully Connected Layer



Softmax



Output Class

Component Roles

Component	Function
Input Layer	Accepts image
Convolution Layer	Extracts low-level features
Residual Blocks	Learn deep features
Skip Connections	Avoid vanishing gradients
Global Pooling	Reduces parameters
Softmax Layer	Final classification

8. Performance Evaluation

Identify the evaluation metrics used in the article, such as:

Accuracy, Precision, Recall, F1-Score , Loss , Mean Squared Error (MSE)

BLEU Score, where applicable

Report the major performance results obtained by the proposed model.

Answer:

Evaluation Metrics

- Accuracy
- Top-1 Error
- Top-5 Error
- Training Loss

Major Results

Model	Top-5 Error
VGGNet	6.8%
GoogLeNet	6.7%
ResNet-152	3.57%

Observation

ResNet-152 achieved the best performance among compared models.

9. Results and Discussion

What are the major findings of the research?

How does the proposed model perform compared with existing methods?

Which model/technique produced the best results?

What factors contributed to the observed performance?

Answer:

Major Findings

- Residual learning significantly improved deep network training.
- ResNet achieved state-of-the-art ImageNet performance.
- Deep networks became easier to optimize.

Comparison with Existing Methods

Model	Performance
AlexNet	Lower
VGGNet	Better
GoogLeNet	Better
ResNet	Best

Factors Contributing to Performance

- Residual learning framework
- Skip connections
- Better gradient flow
- Deep architecture

10. Advantages of the Proposed Model

Identify at least three advantages of the proposed approach.

Answer:

Advantage 1

Improves gradient propagation through skip connections.

Advantage 2

Enables training of very deep neural networks.

Advantage 3

Provides excellent transfer learning capability.

Advantage 4

Reduces degradation problems.

Advantage 5

High classification accuracy.

11. Limitations of the Study

Identify the major limitations mentioned by the authors and any additional limitations you observe, such as: Small dataset, High computational requirements, Training time, Overfitting, Limited generalization, Model complexity.

Answer:

Author Mentioned Limitations

- High computational requirements.
- Large memory consumption.

Additional Limitations

- Long training time.
- Requires GPU resources.
- Complex architecture.
- Not ideal for low-power devices.

12. Critical Analysis

Provide your own analysis of the research article:

Is the selected deep learning architecture appropriate for the problem?

Is the dataset adequate?

Are the evaluation metrics appropriate?

Are the experimental results convincing?

Could another Unit IV technique provide better performance?

Answer:

Is the Architecture Appropriate?

Yes. Residual learning effectively addresses deep network training issues.

Is the Dataset Adequate?

Yes. ImageNet is one of the largest image datasets and provides sufficient diversity.

Are Evaluation Metrics Appropriate?

Yes. Top-1 and Top-5 errors are standard metrics for image classification.

Are Results Convincing?

Yes. Experimental results clearly demonstrate superiority over previous CNN architectures.

Could Another Unit IV Technique Perform Better?

DenseNet could provide stronger feature reuse and fewer parameters. However, ResNet remains simpler and highly effective for transfer learning applications.

13. Future Scope

Suggest possible improvements or future research directions based on the limitations identified in the article.

Answer:

Possible Improvements

1. Combine ResNet with Attention Mechanisms.
2. Use EfficientNet for better efficiency.
3. Reduce computational complexity.
4. Develop lightweight versions for mobile devices.
5. Integrate Vision Transformers (ViT).

14. Conclusion

Provide a brief conclusion summarizing:

The research problem

Proposed model

Major results

Your overall assessment of the article

Answer:

The selected research article addresses the challenge of training very deep neural networks for image classification. The authors proposed the **ResNet architecture**, which introduces residual learning through skip connections. Using the ImageNet dataset, the model achieved a Top-5 error rate of **3.57%**, outperforming previous CNN models. The study demonstrates that residual learning effectively solves degradation problems and improves transfer learning performance. Overall, the article is a highly influential contribution to deep learning and remains widely used in modern computer vision applications.

15. References

Provide the complete citation of the selected research article and any additional references used in the review.

Answer:

1. He, K., Zhang, X., Ren, S., & Sun, J. (2016). *Deep Residual Learning for Image Recognition*. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 770–778.
2. [CVPR Open Access Paper](#)
3. [IEEE Xplore Digital Library](#)
4. [ImageNet Dataset Official Website](#)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Selection & Problem Understanding	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Deep Learning Model & Methodology Analysis	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Results & Performance Evaluation	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Critical Analysis & Technical Evaluation	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Future Scope, Presentation & Documentation	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis